# E-Portfolio

# Table of contents

**Jeff's e-Portfolio** **4**

**1 Introduction** **5**
1.1 About Myself . . . 5
1.2 Overreaching Education/Career Goals . . . 5

**2 Curriculum Vitae** **6**
2.1 **Education and Qualifications** . . . 6
2.2 **Employment History** . . . 7
2.3 **Skills and Main Achievements** . . . 9
2.4 **Selective Interests** . . . 9

**3 Reflection on the MSc Project** **10**
3.0.1 Project Summary . . . 10
3.0.2 Skills Developed . . . 10
3.0.3 Challenges and Solutions . . . 10

**4 Skills Mapping** **12**
4.0.1 Personal Effectiveness . . . 12
4.0.2 Employability . . . 12
4.0.3 Academic Development . . . 12
4.0.4 Research Development . . . 12

**5 SWOT_Analysis** **14**
5.0.1 Strength . . . 14
5.0.2 Weakness . . . 14
5.0.3 Opportunities . . . 15
5.0.4 Threats . . . 15
5.1 SMART Analysis . . . 15

**6 Action Plan** **16**
6.0.1 Primary Objective: Academic Progression . . . 16
6.0.2 Secondary Objective: Professional Industry Experience . . . 16

**7 Career Readiness** **17**
7.0.1 Technical Profile & Market Positioning . . . 17

7.0.2 Academic Pursuits & Professional Development . . . . . 17

**8 Seminar Skill Reflection** **18**

**9 Academic Projects & Assignments** **19**
9.0.1 Mixed-Methods Clinical Data Analysis using MLLMs . . . . . 19
9.1 Extracurricular Activities . . . . . 20
9.1.1 MSc Course Student Representative . . . . . 20
9.2 Visual Showcases & Media . . . . . 22
9.2.1 Research Poster: [LLM for Mixed Method Research in Clinical Trial] . . 22

**10 Summary** **24**

# Jeff's e-Portfolio



Figure 1: This is Jeff

Hi, welcome to Jeff's E-Portfolio.

> **Yesterday is history, tomorrow is a mystery, but today is a gift. That is why it is called the present. — Master Oogway**

# 1 Introduction



## 1.1 About Myself

- Early career scientist with experience in both academic research and clinical trial, aiming to combine wet-lab techniques with computational analysis in academic research.
- Have cycled about everywhere in the world , and an avid guitarist .

## 1.2 Overreaching Education/Career Goals

- Statistical modelling
- Machine Learning programming
- Working with various kinds of data
  - Bioinformatics, -omics data
  - Clinical Trial Data
  - Health Informatics, Electronic Health Record

# 2 Curriculum Vitae

## 2.1 Education and Qualifications

**Natural Science Biological, University of Cambridge, England [2019 - 2022]**

- First Year Modules: Biology of Cells, Physiology of Organisms, Chemistry, Mathematical Biology (Statistics and Programming).
- Second Year Modules: Biochemistry and Molecular Biology, Cell Developmental Biology, Biology of Disease
- Third Year Modules: Genetics (with research project)

**Church of Christ in China Kei Long College, Hong Kong [2013 – 2019]**

- Hong Kong Diploma of Secondary Education (2019):
- Equivalent to A-LEVEL: A*in English Language, A* in Chemistry, A* in Biology, A in Maths, A in Physics, B in Further Maths

**Online Learning – Coursera**

University course completion on topics related to AI, Deep Learning, Bioinformatics, and Programming

- Machine Learning: Theory and Hands-on Practice with Python – University of Colorado Boulder
- Supervised approaches in Machine Learning
- Unsupervised approaches in Machine Learning
- Deep Learning and Neural Networks
- Genomic Data Science – Johns Hopkins
- Python for Genomics
- Command line tools for Genomics
- Bioconductor for Genomics

- Statistics for Genomics
- Applied Data Science with Python – University of Michigan

## 2.2 Employment History

**Clinical Trial Assistant – University College Hospital London [11/2024 – current]**

- Clinical Trial Assistant at the Cancer Clinical Trial Unit at UCLH
- Responsibilities:
  - Facilitate set-up, running, closeout and archiving of cancer clinical trials
  - Administrative tasks, liaising with pharmaceutical sponsors, PIs, patients, hospital departments, to ensure smoothing running of clinics and collection of data.
  - Data management for clinical trial, including patient details, clinical data, recruitment statistics, etc.
  - Perform laboratory procedures (haematology)

**Research Assistant – Northwick Park Institute for Medical Research (The Griffin Institute) [09/2023 – 04/2024]**

- Research assistant at the Contract Research Organization (CRO) department.
- Responsibilities:
  - Plans and delivers bespoke contract pre-clinical trial (animal testing) and report to external clients.
  - Perform lab experiments, data collection and analysis.
  - Maintenance of lab and animal facilities, including restocking supplies.
  - Trained in histology (gross cutting, tissue processing, microtome sectioning, staining, etc.), clinical pathology (haematology testing, blood handling, etc.) and microbiology procedures (antiseptic techniques, etc.), animal facility maintenance (tracking humidity, etc.).
  - Experience working in a GLP lab (drafting SOP, system validations, etc.)

**Final Year Research Project Placement, Summer's Lab, Department of Genetics, University of Cambridge [02/2022 – 04/2022]**

- Project on growth relationship between *E. coli* and *Lactobacillus* in human gut microbiome

- Investigated growth of *E. coli* in different culture conditions: growth culture supplemented with different products of *E. coli* and *Lactobacillus* tryptophan metabolism. Investigated growth of different strains *E. coli* and *Lactobacillus* when cultured together. Bacterial growth tracked in CFU counts over 48 hours.
- Trained in microbiology. Hands on experience with aseptic technique, bacterial culture, colony forming units (CFU) counts, serial dilution, operating plate reader and O.D measurements, histology techniques (preparing slides, gram staining)
- Experience working with genetically modified bacteria, adhering to high standard of laboratory safety regulations. Precision and attention to details.

**Summer Intern, Luisi's Lab, Department of Biochemistry, University of Cambridge [08/2021 – 10/2021]**

- Project on protein purification and structural identification of Ribonuclease E.
  - Investigated different approaches to isolate RNase E involving his tag pull down, flag tag pull down, and gel filtration chromatography.
  - Investigated if RNase E interacts with polysome complex in cells, by using YFP fluorescent gel filtration chromatography.
  - Investigated if RNase E interact with NUS transcription factors, by carrying out NUS-his tag pull down and western blots.
  - Hands on experience with general biochemical techniques like gel electrophoresis, western blots, bacterial cultivation, PCR, antibody pull-down assays. Also, learnt specific techniques used in structural biology like HPLC, cryo-EM, qPCR, etc.
  - Review writing, navigating secondary scientific database, researching scientific literature.

**President [2021-2022], Transport Secretary [2020 – 2021],**

**Cambridge University Hill Walking Club**

- Duties include leading the club, delegate tasks and ensure smooth running of the club. Act as representative for the club, communicating with other university clubs and external companies.
- Communication and interpersonal skills, teamwork, leading and collaborating with committee members.

## 2.3 Skills and Main Achievements

**Programming and Data Analysis**

· Proficient in R (Mathematical Biology module, Natural Science, Cambridge) and Python (Self Taught. Coursera.)

· Proficient in python data analysis libraries like numpy, pandas, scipy, matplotlib.

· Fluent in building supervised machine learning models in python (sklearn library). Fluent in building deep Neural Network models in python (pytorch).

**Australian National Chemistry Quiz – highest distinction attained *(Class Excellence)*** [2017]

**Secondary School Mathematics and Science Competition by Polytechnic University of Hong Kong – top 15% in Biology, Chemistry** [2018]

## 2.4 Selective Interests

- Marathon/Trail Runner. Finisher of 2018 HK50 25km Trail Race.
- Cycled the entirety of the U.K by bicycle. (Lands' End to John O'Groats)
- Cycled from London to Rome

# 3 Reflection on the MSc Project

### 3.0.1 Project Summary

My MSc project evaluated the feasibility of using locally hosted Multimodal Large Language Models (MLLMs) to computationally integrate and analyse both qualitative and quantitative clinical trial data. Using a dataset from the LQD trial for treatment-resistant depression, I designed and applied three mixed-method analytical tasks: Convergence, Explanatory, and Statistical. Because a human expert baseline was unavailable, I developed an automated "LLM-as-a-Judge" evaluation strategy using custom scoring rubrics to assess the model's output. The project successfully demonstrated that MLLMs exhibit strong logical reasoning and factual consistency when triangulating clinical data, providing a foundational framework for future computational clinical analysis.

### 3.0.2 Skills Developed

Throughout the research process, I developed advanced technical data analysis and project management skills, specifically by deploying and configuring language models on a High-Performance Cluster (CREATE). I also significantly enhanced my critical thinking and problem-solving abilities by designing multi-stage "chain-of-thought" prompting pipelines and systematically evaluating the complex logical relationships generated by the AI.

### 3.0.3 Challenges and Solutions

| Challenge | How I Overcame It |
|---|---|
| **Out-Of-Memory (OOM) Errors:** Limited computing resources on the HPC caused frequent memory crashes during analysis. | I implemented model quantization to load weights in 4-bit precision, utilized memory-efficient scaled dot-product attention, and divided massive qualitative transcripts into smaller, memory-safe batches. |
| **Clinical Misinterpretation:** The model incorrectly assumed a lower QIDS-SR score indicated more severe depression symptoms. | I refined the prompt engineering by adding targeted system prompts that explicitly provided the correct clinical context and scale interpretation. |

| | |
|---|---|
| **Data Pipeline Failures:** The LLM-generated Python code failed to locate specific data columns in the unseen quantitative dataset. | I updated the prompt instructions to include a detailed glossary of variable names, ensuring the model had the necessary context to generate functional code. |

# 4 Skills Mapping

### 4.0.1 Personal Effectiveness

- **Time Management & Resilience:** Successfully balanced demanding project milestones alongside coursework, demonstrating resilience by adapting to technical setbacks such as computational limits and coding challenges.
- **Adaptability & Leadership:** Took ownership of complex problem-solving processes, pivoting research strategies when initial methodologies required refinement.

### 4.0.2 Employability

- **Professional Skills:** Developed highly transferable technical competencies in data analysis, Python programming, and AI model deployment.
- **Networking & Applications:** Actively collaborated with e-research teams and academic peers, building a concrete portfolio of evidence to support future job applications and technical interviews.

### 4.0.3 Academic Development

- **Learning Progress:** Transitioned from foundational module concepts to executing advanced, self-directed research, showcasing a steep and continuous learning curve.
- **Module Reflections & Assessments:** Consistently applied theoretical frameworks from coursework into practical, graded assessments, using feedback to iteratively improve academic performance.

### 4.0.4 Research Development

- **Methods & Ethics:** Designed robust mixed-methods research frameworks while strictly adhering to data privacy protocols and ethical guidelines for handling sensitive clinical data.
- **Dissemination:** Effectively communicated complex statistical findings and technical methodologies through structured academic writing and comprehensive reporting.

#### 4.0.5

# 5 SWOT_Analysis

### 5.0.1 Strength

- Academic Soft Skill
  - Critical Thinking
  - Problem solving
- Academic Hard Skill
  - Range of wet lab techniques
  - Clinical Trial knowledge
  - Coding skill for data analysis
  - Knowledge on Machine Learning Systems
- General Skill
  - Public speaking and presentation skills
  - Leadership skills
  - Communication skill

### 5.0.2 Weakness

- Academic Hard Skill
  - Mathematical intuitions behind Machine Learning concepts
  - Computer science concepts behind algorithms and software
  - How different research fields connect with each other
- General Skills
  - Multi-tasking
  - Teamwork (works better alone)

### 5.0.3 Opportunities

- MSc Applied Statistical Modelling and Health Informatics
- Opportunity to speak to/work with researchers in machine learning in healthcare field
- Opportunity to access large amount of healthcare data for research insight

### 5.0.4 Threats

- A busy schedule with MSc, while balancing social life, can mean little time to do extra work reaching out to professors or extra work within the curriculum.
- Applying for PhD is very time consuming, which requires balancing MSc workload and other commitments.

## 5.1 SMART Analysis

Specific, Measuerable, Achieveable, Relevant, Time-bound

1. Successfully complete all modules in the MSc course and comprehend all material.
2. Achieve Merit/Distinction in the MSc course
3. By September 2026, I would have either
    - Enrolled in a fully funded PhD program
    - In full time employment, in a field related to data analysis
4. Aim to publish the report from my Master dissertation

# 6 Action Plan

### 6.0.1 Primary Objective: Academic Progression

- **Pursue Doctoral Studies:** My principal career objective remains advancing within academia by identifying and securing a suitable PhD opportunity.
- **Enhance Funding Competitiveness:** Following consultations with academic staff, I have identified that acquiring post-MSc Research Assistant (RA) experience is critical for becoming a highly competitive candidate for PhD funding.
- **Secure Departmental Research Roles:** To bridge this experience gap, I am actively seeking collaborative academic research positions within the department to work alongside established academics in my field.

### 6.0.2 Secondary Objective: Professional Industry Experience

- **Develop Applied Data Science Skills:** As a strategic alternative, I am concurrently exploring professional employment opportunities within the healthcare sector.
- **Bridge Academia and Industry:** Securing a role in industry will allow me to hone my healthcare data science skills in a practical, real-world setting, which directly supports my ultimate long-term career ambitions following the completion of a PhD.

# 7 Career Readiness

### 7.0.1 Technical Profile & Market Positioning

- **Comprehensive Skill Foundation:** Throughout my MSc, I have built a robust and diverse technical repertoire encompassing statistics, data science, machine learning, natural language processing, and advanced Large Language Model (LLM) applications.
- **Strategic Market Entry:** While my academic breadth is extensive, I recognize that I am currently a generalist rather than an industry specialist. To bridge this gap and develop highly specific, industry-ready expertise, I am strategically targeting entry-level to lower-mid-level professional roles.

### 7.0.2 Academic Pursuits & Professional Development

- **Doctoral Application Experience:** I proactively submitted PhD applications throughout the academic year and successfully passed the rigorous departmental PhD interview. Although I was not nominated for funding this cycle due to limited availability and high competition, the process provided invaluable experience.
- **Career Strategy & Networking:** To continuously refine my trajectory, I have engaged extensively with university career services to optimize my future PhD applications and enhance my networking strategies for the industry job hunt.

# 8 Seminar Skill Reflection

- **GitHub and Version Control:** The most useful professional development session I attended was the GitHub skills seminar.
- **Technical Understanding:** This session significantly deepened my practical knowledge, helping me truly understand Git as a comprehensive version control system.
- **Broader Application:** Furthermore, it allowed me to fundamentally reframe my perspective on GitHub, recognizing its capabilities as much more than just a basic repository for sharing code.

# 9 Academic Projects & Assignments

### 9.0.1 Mixed-Methods Clinical Data Analysis using MLLMs

- **Overview:**
  - This project evaluated the feasibility of utilizing locally hosted Multimodal Large Language Models (MLLMs) to computationally integrate and jointly analyze qualitative and quantitative clinical trial data.
  - Using a dataset from the LQD trial for treatment-resistant depression, three mixed-method analytical tasks—Convergence, Explanatory, and Statistical—were applied to the data using an open-source MLLM.
  - Due to the absence of a definitive human ground truth, an automated "LLM-as-a-Judge" evaluation strategy was implemented to systematically assess the model's analytical performance.
- **Key Findings:**
  - The MLLM demonstrated strong overall performance across the mixed-method tasks, achieving high factual consistency and effectively generating functional Python code for quantitative data analysis.
  - The model successfully analyzed complex logical relationships, effectively bridging qualitative patient narratives with quantitative data trajectories.
  - While the model showed robust reasoning abilities, a primary limitation observed was its inherent lack of deep, domain-specific clinical knowledge, which required targeted prompting and context to overcome.
- **My Contribution:**
  - I designed the three mixed-method research tasks and developed the multi-stage "chain-of-thought" prompting pipelines used to guide the model's analysis.
  - I created the specific scoring rubrics utilized by the independent "LLM-as-a-Judge" to evaluate the model's factual consistency, coding robustness, and integrative synthesis.

– I managed the technical deployment of the models on King's College London's High Performance Cluster (CREATE), utilizing technical strategies like model quantization in 4-bit precision and data batching to successfully overcome Out-Of-Memory (OOM) errors.

---

## 9.1 Extracurricular Activities

### 9.1.1 MSc Course Student Representative

- **Role:** Attend departmental meeting, reflect student opinions on the course.

---

## 9.2 Visual Showcases & Media

### 9.2.1 Research Poster: [LLM for Mixed Method Research in Clinical Trial]

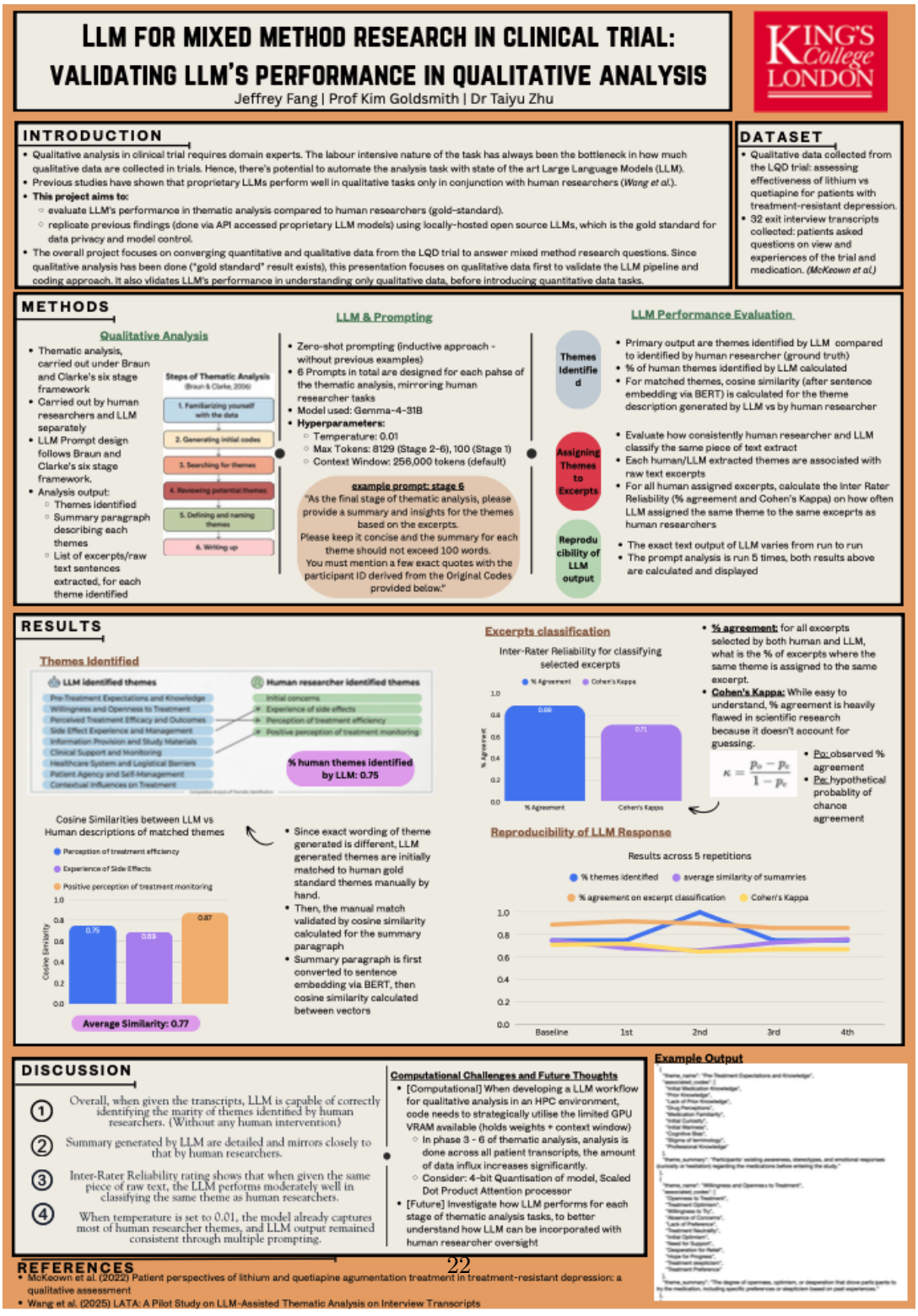



Figure 9.1: Research Poster

## 9.3

# 10 Summary

Reflecting on my MSc journey, bringing together my coursework, technical skill development, and the execution of my dissertation on clinical data integration, I can clearly see my transformation from a student into a capable researcher[cite: 24]. *In hindsight*, the most crucial advice I would give myself at the start of the programme would be to embrace the steep technical learning curve early on and to initiate conversations regarding PhD and career pathways from day one[cite: 24]. Regarding *missing skills*, I quickly realized that while I grasped the theoretical aspects of health data, I initially lacked the advanced computational proficiencies—such as managing High-Performance Clusters (HPC), optimizing Python environments, and utilizing robust version control like Git—that are essential for executing complex, large-scale AI research[cite: 24]. Looking toward *future growth*, my immediate next steps involve securing post-MSc Research Assistant experience to become a more competitive candidate for PhD funding, while simultaneously continuing to hone my applied healthcare data science skills to ensure I am highly prepared for both academic research and professional industry roles.